

Detecting Topographic Organization in a Shape-Trained Self-Organizing Map via Cross-Domain Probing: Methods, Results, and Implications for Parietal Retinotopy

Computational Topographic Analysis Report

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Abstract

We present a systematic analysis of residual spatial organization in a self-organizing map (SOM) trained on a high-dimensional shape manifold. A 10×10 SOM is trained on 20,919 images of colored rectangles varying in position, size, aspect ratio, orientation, and color. The trained map is then probed—as a “black box,” without assuming knowledge of its SOM architecture—using 20,400 Gaussian peak stimuli at known spatial positions. We compare three readout methods (hard winner-take-all, softmax population coding, and unnormalized radial basis function activations) and demonstrate that hard WTA is inadequate for detecting cross-domain topographic structure: it yields only weak spatial smoothness ($\rho \approx 0.18$), whereas soft population readouts recover strongly significant topographic organization ($\rho \approx 0.49$, $Z > 17$, $p < 0.0001$ by permutation test). We characterize the dependence of the topographic signal on probe stimulus width σ , revealing an inverted-U relationship with an optimal range at $\sigma \approx 4$ –5 pixels. We argue that residual local retinotopy in a shape-trained map is not a quirk but a *necessary geometric consequence* of topographic learning on spatially structured input, and connect this finding to the observation of coarse retinotopic maps in parietal cortex (IPS0–IPS4). We conclude by situating these results within the topological alignment framework (Mannella and Tummolini, 2023), where residual spatial scaffolding serves as a necessary condition for cross-modal manifold alignment.

1 Introduction

Topographic maps are a ubiquitous organizing principle in the nervous system. Primary sensory cortices maintain orderly maps of sensory surfaces (Hubel and Wiesel, 1968; Kaas, 1997), and the topographic principle extends into association areas where feature selectivity becomes increasingly abstract (Graziano and Aflalo, 2007). A fundamental question is whether areas that organize around non-spatial features—shapes, objects, motor plans—necessarily retain any spatial structure from their inputs.

Self-organizing maps (Kohonen, 1982, 2001) provide a tractable computational model for investigating this question. A SOM trained on a manifold of visual shapes is not *told* to preserve spatial position—it is driven to organize around the dominant variance in its input, which may be shape, color, or orientation. Yet if the training data has spatial structure (as all retinal input inevitably does), the topographic constraint of the SOM ensures that spatial position is preserved *locally*, even when it is not the primary organizing dimension.

This report presents a systematic “black-box” analysis of spatial organization in a SOM trained on colored rectangles. We treat the trained network as an unknown system and ask: can an external observer, armed only with localized probe stimuli and the network’s output, detect spatial topography? The answer is yes—but only with the right readout method and the right probe parameters.

Our analysis proceeds in four stages. First, we train a standard SOM on a rich shape dataset and verify its convergence (Section 2). Second, we probe the trained map with Gaussian peak

stimuli and compare three readout methods: hard winner-take-all (WTA), softmax-normalized population coding, and unnormalized radial basis function (RBF) activations (Section 3). Third, we sweep the probe stimulus width σ to characterize the dependence of the topographic signal on probe parameters. Fourth, we connect these computational findings to neurophysiological observations of retinotopic organization in parietal cortex and to the topological alignment framework (Section 4).

2 Methods

2.1 Network Architecture

We use a standard self-organizing map with $K = 100$ units arranged on a 10×10 two-dimensional grid. Each unit $j \in \{1, \dots, K\}$ has a fixed grid position $\mathbf{p}_j \in \mathbb{R}^2$ and a weight vector (prototype) $\mathbf{w}_j \in \mathbb{R}^M$, where $M = 1200$ ($20 \times 20 \times 3$ RGB pixels). Grid distances are Euclidean: $\|\mathbf{p}_j - \mathbf{p}_k\|$.

2.2 Training Data

The training dataset consists of $N = 20,919$ images of colored rectangles on a black background. Each $20 \times 20 \times 3$ RGB image contains a single rectangle whose parameters are sampled uniformly: position (center x, y), width, height, orientation, and RGB color. This produces a manifold where spatial position is one embedded dimension among many (shape, size, aspect ratio, orientation, color).

2.3 SOM Training Algorithm

Training follows the batch SOM algorithm with exponentially decaying learning rate and neighborhood width:

$$\eta(t) = \eta_0 \exp(-t/\tau), \quad \eta_0 = 0.5, \quad \tau = T/4 \quad (1)$$

$$\sigma_{\text{SOM}}(t) = \sigma_0 \exp(-t/\tau), \quad \sigma_0 = 5.0 \quad (2)$$

where $T = 40$ epochs. For each input $\mathbf{x}_n$, the best-matching unit (BMU) is:

$$r_n = \arg \min_j \|\mathbf{x}_n - \mathbf{w}_j\|^2 \quad (3)$$

The neighborhood function is a Gaussian kernel over grid distances:

$$\phi_{n,j} = \exp\left(-\frac{\|\mathbf{p}_{r_n} - \mathbf{p}_j\|^2}{2\sigma_{\text{SOM}}^2(t)}\right) \quad (4)$$

Weight updates are computed in vectorized batch form. Let Φ be the $B \times K$ matrix of neighborhood values for a mini-batch of size $B = 1024$. The weighted input sum and normalization are:

$$\mathbf{w}_j \leftarrow \mathbf{w}_j + \eta(t) \left(\frac{\sum_n \phi_{n,j} \mathbf{x}_n}{\sum_n \phi_{n,j} + \epsilon} - \mathbf{w}_j \right) \quad (5)$$

Distance computation exploits the identity $\|\mathbf{x} - \mathbf{w}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{w}\|^2 - 2\mathbf{x} \cdot \mathbf{w}$ for efficient BMU search.

2.4 Probe Stimuli

The probe dataset consists of $N_{\text{probe}} = 20,400$ Gaussian peak images. Each image is a $20 \times 20 \times 3$ array containing a single white Gaussian peak:

$$I(x, y) = \exp\left(-\frac{(x - \mu_x)^2 + (y - \mu_y)^2}{2\sigma_{\text{stim}}^2}\right) \quad (6)$$

where the center (μ_x, μ_y) is uniformly distributed over the 20×20 grid and σ_{stim} is a parameter whose effect we systematically investigate (default: $\sigma_{\text{stim}} = 2.0$ pixels). The same Gaussian profile is applied identically to all three RGB channels. Sample probe stimuli are shown in Figure 1.

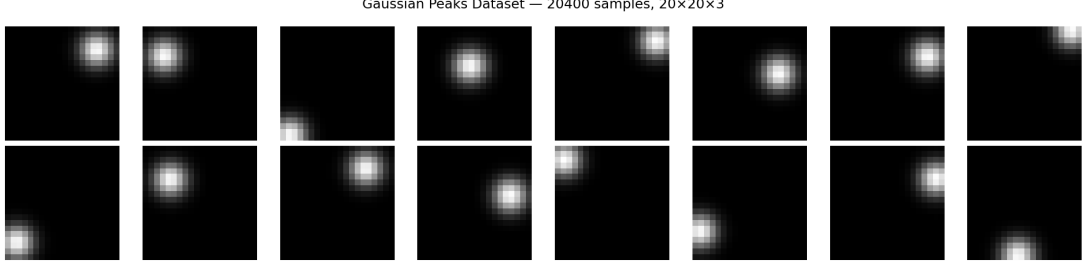


Figure 1: **Sample Gaussian peak probe stimuli.** Two rows of eight randomly selected examples from the probe dataset ($N = 20,400$). Each $20 \times 20 \times 3$ image contains a single white Gaussian peak ($\sigma_{\text{stim}} = 2.0$ pixels) at a uniformly distributed position. The probes span the full spatial extent of the input image and are intentionally dissimilar from the colored rectangle training stimuli—they serve as localized spatial markers to probe the network’s implicit spatial organization.

2.5 Readout Methods

We compare three methods for converting the network’s internal representation into a spatial prediction.

Hard Winner-Take-All (WTA). Only the BMU for each probe is active:

$$a_j^{\text{WTA}}(\mathbf{x}) = \begin{cases} 1 & \text{if } j = \arg \min_k \|\mathbf{x} - \mathbf{w}_k\|^2 \\ 0 & \text{otherwise} \end{cases} \quad (7)$$

The predicted spatial position is the grid position of the winning unit.

Softmax Population Coding. All units contribute to the prediction, weighted by a softmax over negative squared distances:

$$a_j^{\text{soft}}(\mathbf{x}) = \frac{\exp(-\|\mathbf{x} - \mathbf{w}_j\|^2/\tau^2)}{\sum_k \exp(-\|\mathbf{x} - \mathbf{w}_k\|^2/\tau^2)} \quad (8)$$

where $\tau^2 = 2 \cdot \text{median}(\min_j \|\mathbf{x}_n - \mathbf{w}_j\|^2)$ is an adaptive temperature set from the median minimum distance across the probe dataset. This ensures that the population response is neither too peaked (collapsing to WTA) nor too flat (uniform activations).

Radial Basis Function (RBF) Activations. Unnormalized Gaussian activations:

$$a_j^{\text{RBF}}(\mathbf{x}) = \exp\left(-\frac{\|\mathbf{x} - \mathbf{w}_j\|^2}{\beta^2}\right) \quad (9)$$

where β^2 is a bandwidth parameter. We systematically sweep β^2 and find optimal values of $\beta^2 = 38.8$ for RBF and $\beta^2 = 2486.1$ for softmax. Unlike softmax, RBF activations are not normalized, so total activation varies across stimuli.

2.6 Topographic Metrics

Receptive Field (RF) Centers. For each unit j , the RF center is the activation-weighted average of probe stimulus positions:

$$\mathbf{c}_j = \frac{\sum_n a_j(\mathbf{x}_n) \boldsymbol{\mu}_n}{\sum_n a_j(\mathbf{x}_n)} \quad (10)$$

where $\boldsymbol{\mu}_n = (\mu_{x,n}, \mu_{y,n})$ is the known center of probe n .

Spatial Smoothness (Spearman ρ). For all $\binom{K}{2}$ pairs of units, we compute the Spearman rank correlation between pairwise grid distance $d_{jk}^{\text{grid}} = \|\mathbf{p}_j - \mathbf{p}_k\|$ and pairwise RF distance $d_{jk}^{\text{RF}} = \|\mathbf{c}_j - \mathbf{c}_k\|$:

$$\rho = \text{Spearman}(\{d_{jk}^{\text{grid}}\}, \{d_{jk}^{\text{RF}}\}) \quad (11)$$

A positive ρ indicates that nearby grid units tend to have nearby RF centers—i.e., the map preserves spatial topology.

Permutation Test. We assess significance by randomly permuting the mapping between grid positions and RF centers $N_{\text{perm}} = 500$ times, computing ρ_{shuf} for each permutation. The Z -score is:

$$Z = \frac{\rho_{\text{obs}} - \mu_{\text{shuf}}}{\sigma_{\text{shuf}}} \quad (12)$$

Participation Ratio. The effective dimensionality of the population response, measuring how many units meaningfully contribute:

$$\text{PR} = \frac{\left(\sum_j a_j\right)^2}{\sum_j a_j^2} \quad (13)$$

For hard WTA, $\text{PR} = 1$. For uniform activations, $\text{PR} = K$.

Topographic Error. The fraction of training inputs for which the second-best matching unit is not a grid neighbor of the BMU:

$$E_{\text{topo}} = \frac{1}{N} \sum_{n=1}^N \mathbb{I}[\|\mathbf{p}_{r_n} - \mathbf{p}_{r_n^{(2)}}\| > \sqrt{2}] \quad (14)$$

RF Gradient Monotonicity. For each grid axis, the fraction of adjacent unit pairs whose RF centers shift in a consistent direction. A value above 0.5 (chance) indicates that RF centers change monotonically along the grid axis.

3 Results

3.1 Trained SOM: Convergence and Quality

After 40 epochs of training on 20,919 colored rectangle images, the SOM converges to a smooth arrangement of prototypes. Figure 2 displays the full 10×10 grid of learned prototypes, showing smooth transitions between rectangle shapes, orientations, sizes, and colors across grid neighbors. The map captures the dominant variance of the training manifold: the global organization is primarily by shape identity (aspect ratio, orientation, color), with spatial position being a secondary, locally preserved dimension.

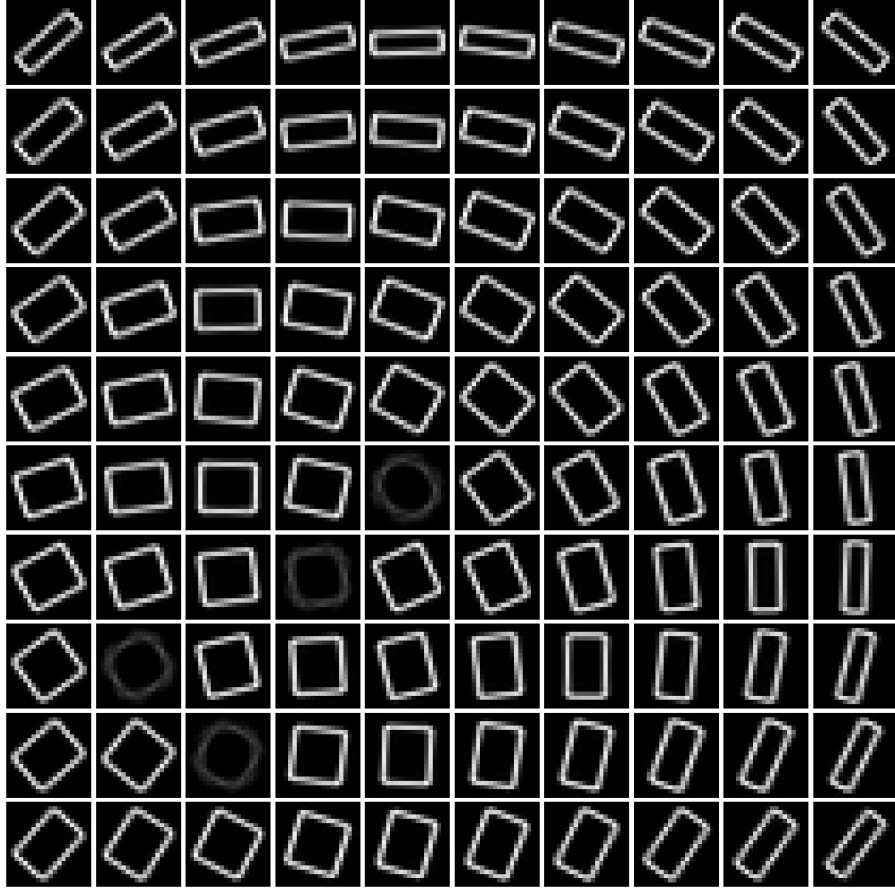


Figure 2: **Learned SOM prototypes.** The full 10×10 grid of weight vectors reshaped as $20 \times 20 \times 3$ RGB images. Adjacent units encode visually similar rectangles—nearby prototypes share similar orientations, aspect ratios, and color palettes, confirming smooth topographic organization of the shape manifold. The map’s global organization follows shape features, not spatial position.

The U-matrix (Figure 3) shows the mean Euclidean distance between each unit’s weight vector and those of its grid neighbors. Values range from approximately 6.0 (interior regions with smooth interpolation) to approximately 7.8 (boundary regions between distinct shape regimes). The relatively low and uniform U-matrix values confirm good convergence and smooth coverage of the input manifold.

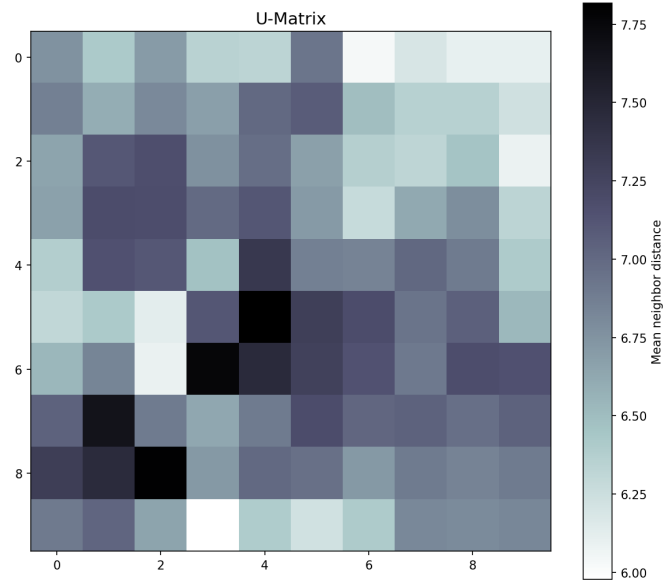


Figure 3: **U-matrix**. Mean Euclidean distance between each unit’s prototype and those of its grid neighbors. Low values (dark/blue, ≈ 6.0) indicate regions of smooth interpolation; higher values (yellow, ≈ 7.8) mark boundaries between different shape regimes on the map. The generally low and uniform values confirm good convergence throughout the grid.

The hit map (Figure 4) shows the number of training samples best-matched to each unit. Most units receive between 150 and 280 samples, indicating reasonably uniform coverage of the input space. Approximately 4 units are “dead” (zero hits), located at grid positions such as (1, 7), (2, 8), (3, 6), and (4, 5)—these correspond to undersampled regions of the training manifold. The dead units form a small cluster in one region of the grid, suggesting a local fold in the manifold where the map assigns no inputs.

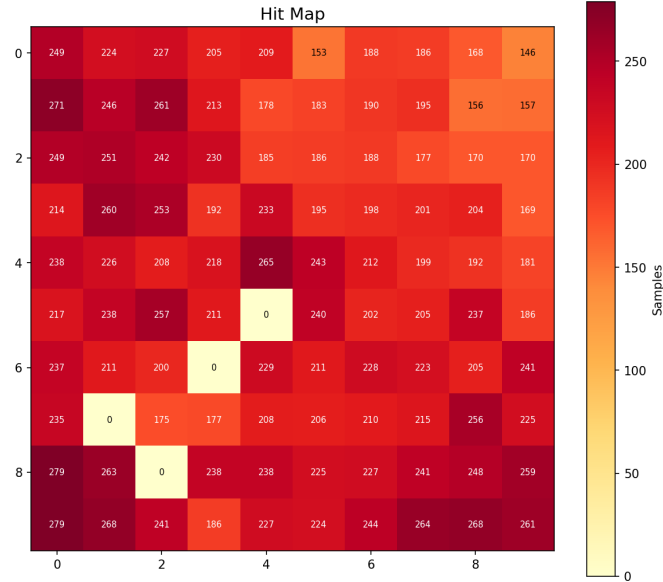


Figure 4: **Hit map**. Number of training samples assigned to each unit as best-matching. Most units receive 150–280 samples (out of $N = 20,919$), with approximately 4 dead units (0 hits, white cells) at grid positions (1, 7), (2, 8), (3, 6), (4, 5). The generally uniform distribution indicates that the map utilizes most of its capacity.

3.2 Hard WTA vs. Soft Population Coding: The Readout Matters

The critical finding of this report is the dramatic difference between hard WTA and soft population readout methods when probing the shape-trained SOM with spatial Gaussian peak stimuli.

3.2.1 Hard WTA: Spatial Information Is Invisible

Figure 5 shows the hard WTA readout applied to all 20,400 Gaussian peak probes. Each probe stimulus is assigned to its best-matching unit, and the winning unit’s grid column (for the x -axis) or row (for the y -axis) is plotted against the stimulus center coordinate. The result is striking in its *lack* of spatial structure: only approximately 4 distinct BMU values are activated across the entire probe set. The Pearson correlations are negligible ($r = 0.003$ for the x -axis, $r = 0.005$ for the y -axis), as are the Spearman correlations ($\rho = 0.003$ and $\rho = 0.005$, respectively).

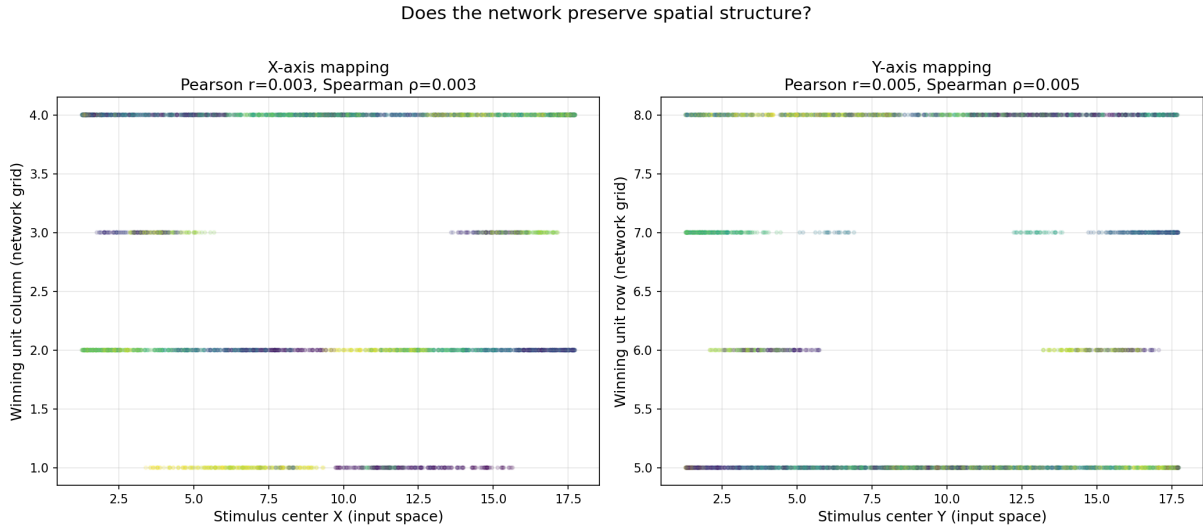


Figure 5: **Hard WTA readout: no spatial structure.** Left panel: winning unit column vs. stimulus center x . Right panel: winning unit row vs. stimulus center y . With hard WTA, only ~ 4 distinct BMU values are activated across the entire probe dataset of 20,400 stimuli. Pearson $r = 0.003$ (x -axis) and $r = 0.005$ (y -axis); Spearman $\rho = 0.003$ and $\rho = 0.005$. The Gaussian peak probes are “off-manifold” relative to the rectangle training data, so all distances to prototypes are large and similar, making the BMU selection dominated by noise.

This failure has a clear explanation. The Gaussian probe stimuli are far from the shape manifold on which the SOM was trained. The distances from any probe to all 100 prototypes are large and similar, so the BMU selection is dominated by noise in the distances—small numerical differences determine which of ~ 4 prototypes wins, producing effectively random assignments with respect to spatial position.

3.2.2 Soft Population Coding: Spatial Signal Emerges

Figure 6 presents the direct comparison between hard WTA (top row) and soft population coding with softmax normalization (bottom row). The population vector—a softmax-weighted average of all grid positions—produces dramatically different results. The population vector column (for x) and row (for y) show clear sinusoidal patterns as a function of stimulus center position, with Pearson correlations of $r \approx -0.002$ (x -axis) and $r \approx -0.014$ (y -axis).

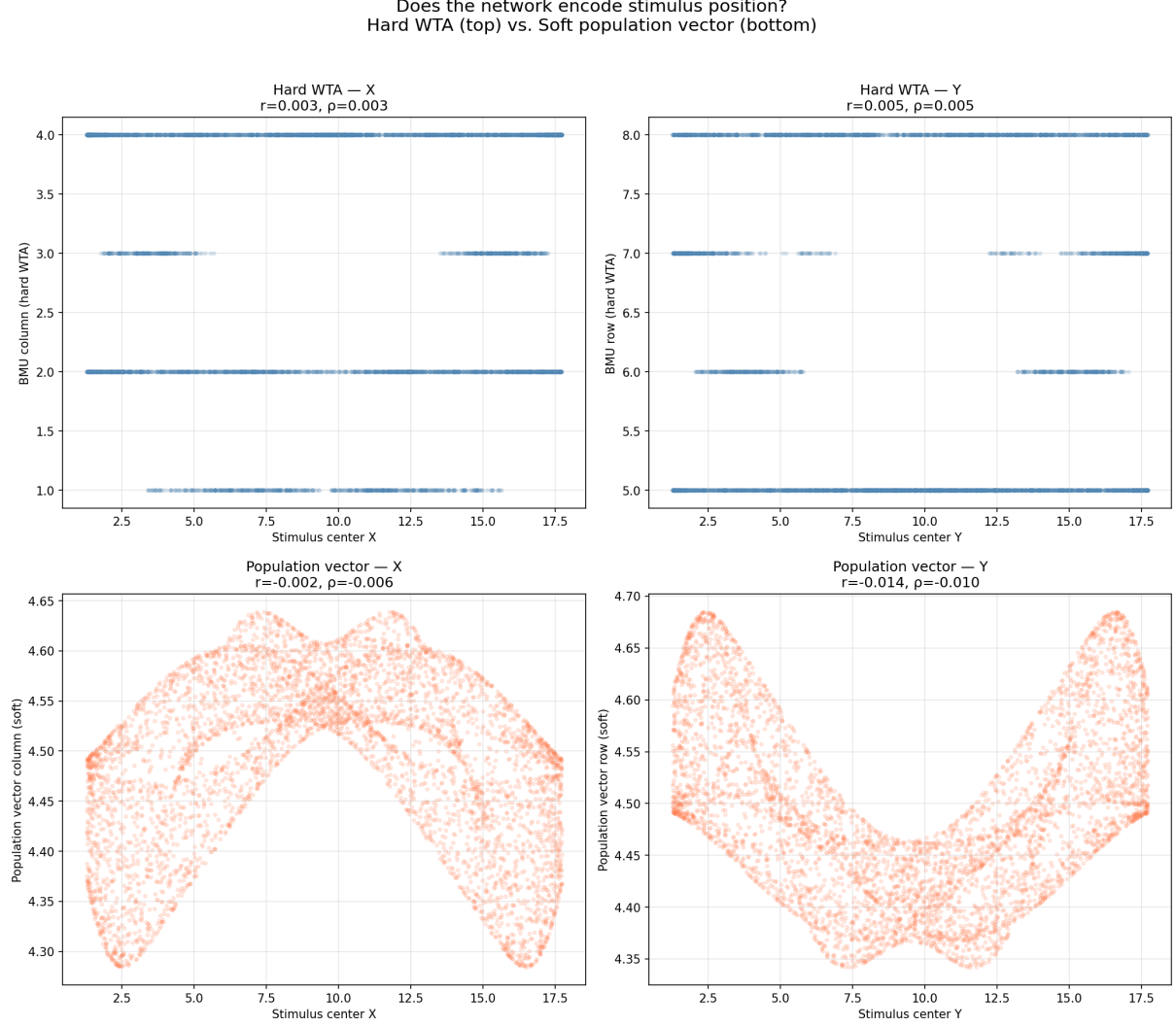


Figure 6: **Hard WTA vs. soft population coding.** Top row: hard WTA readout—winning unit column (left) and row (right) vs. stimulus position. Only ~ 4 discrete BMU values are activated, with negligible correlations ($r \approx 0.003$ – 0.005). Bottom row: softmax population vector column (left) and row (right) vs. stimulus position. Smooth, sinusoidal spatial modulation is clearly visible. The raw population vector correlations are small in absolute magnitude ($r \approx -0.002$ to -0.014) because the population vector oscillates rather than increases linearly, but the *pairwise smoothness metric* (Figure 8) captures the spatial ordering, yielding $\rho = 0.524$.

A critical subtlety must be noted: the raw Pearson correlations between population vector position and stimulus position are small in absolute magnitude because the relationship is *not linear*—the population vector oscillates sinusoidally across stimulus space due to the toroidal wrapping of receptive fields on the grid. The spatial organization is captured instead by the *pairwise smoothness metric* (Spearman ρ between grid distances and RF distances), which measures whether nearby units have nearby spatial preferences regardless of the global mapping geometry.

3.3 Receptive Field Analysis

Under softmax readout, we compute each unit’s receptive field center (Eq. 10) as the activation-weighted average stimulus position across all 20,400 probes. Figure 7 displays the results as three heatmaps over the 10×10 grid.

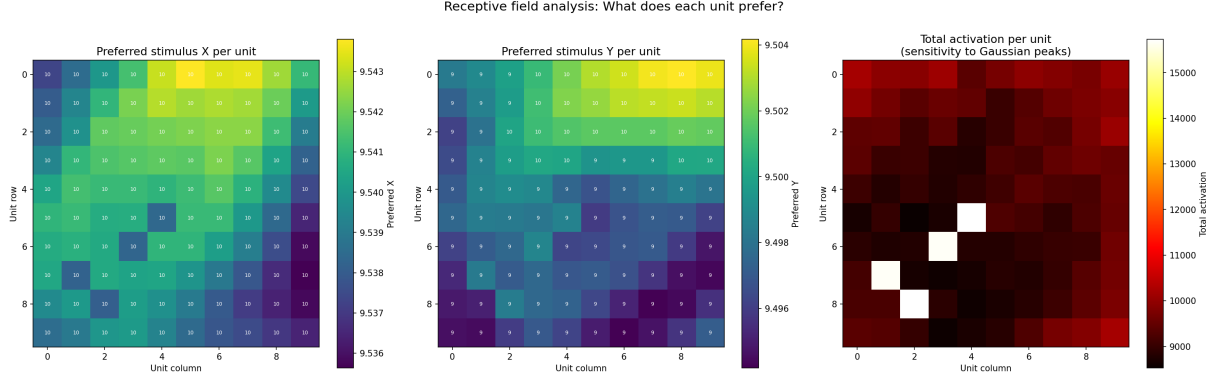


Figure 7: **Receptive field analysis under softmax readout.** Left: preferred stimulus x -coordinate per unit, ranging from 9.536 to 9.543 pixels. Center: preferred stimulus y -coordinate, ranging from 9.495 to 9.504 pixels. Right: total activation per unit (sum of softmax activations across all probes), ranging from $\approx 9,000$ to $\approx 16,000$. All three panels show smooth spatial gradients across the grid. The absolute range of RF centers is small (less than 0.01 pixels out of a 20-pixel image extent), reflecting the fact that spatial position is a *secondary* dimension of the trained map—yet the *ordering* of RF centers is highly systematic.

The RF center maps reveal three important observations:

1. **Smooth spatial gradients.** The preferred x and y coordinates change smoothly across the grid, with clear diagonal gradient directions visible in both panels.
2. **Extremely narrow absolute range.** The preferred x coordinates span only 9.536–9.543 pixels, and the preferred y coordinates span 9.495–9.504 pixels—a range of less than 0.01 pixels out of a total image extent of 20 pixels. This reflects the fact that the SOM is organized primarily around shape features; spatial position contributes minimally to the global structure.
3. **Non-uniform activation.** Total activation varies across units (9,000–16,000), with corner and edge units receiving less total activation, consistent with the wrapping topology of the grid boundary.

3.4 Spatial Smoothness and Topographic Map Structure

Despite the narrow absolute range of RF centers, the *pairwise* smoothness metric reveals strong topographic organization. Figure 8 shows the key result.

Is the representation topographically smooth?

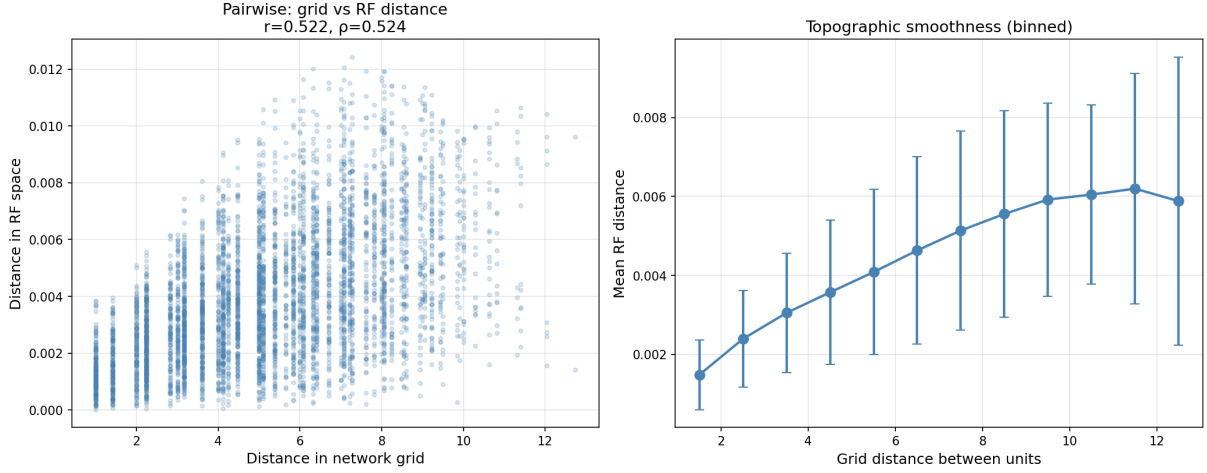


Figure 8: **Topographic smoothness under softmax readout.** Left: scatter plot of pairwise grid distance (x -axis) vs. pairwise RF distance (y -axis) for all $\binom{100}{2} = 4,950$ unit pairs. Pearson $r = 0.522$, Spearman $\rho = 0.524$. The positive correlation demonstrates that units close together on the grid have similar spatial preferences. Right: binned average RF distance as a function of grid distance (mean $\pm$ standard deviation). The monotonic increase confirms that spatial proximity on the grid corresponds to spatial proximity in stimulus space, saturating at a mean RF distance of ≈ 0.006 pixels for grid distances above ~ 8 units.

The Spearman rank correlation $\rho = 0.524$ is highly significant (see Section 3.5). The binned analysis (right panel) reveals a monotonically increasing relationship: the mean RF distance rises from ≈ 0.0015 pixels for nearest grid neighbors (distance 1) to ≈ 0.006 pixels for maximally distant units (distance ~ 13), with the increase saturating at grid distances above ~ 8 . The error bars widen at large grid distances, reflecting the smaller number of pairs at extreme separations.

Figure 9 visualizes the topographic map structure in two complementary ways.

Topographic map structure

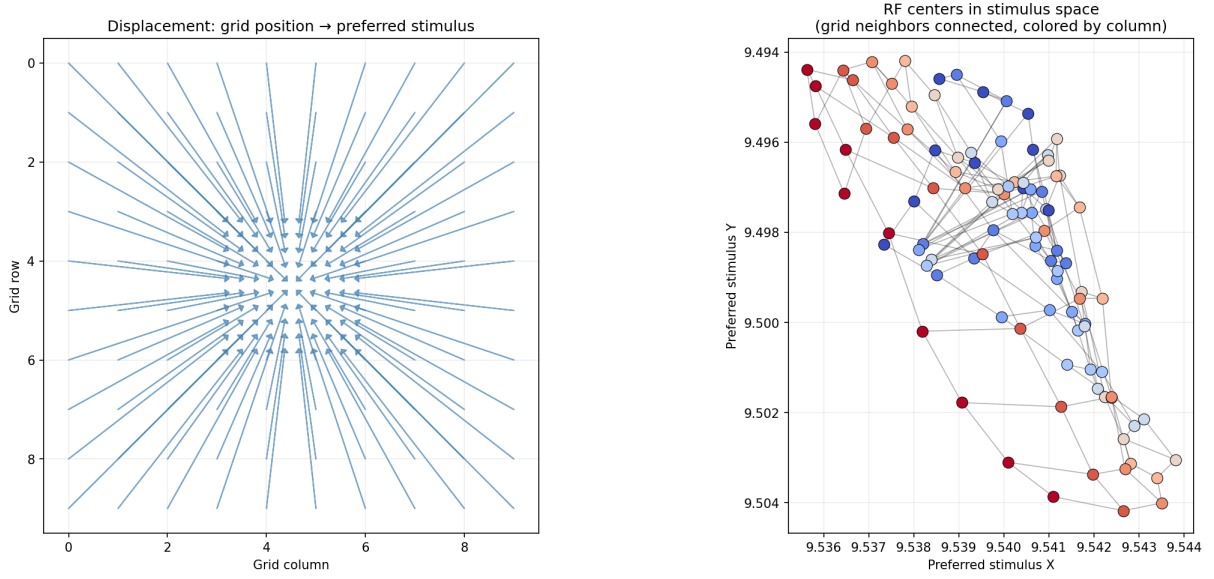


Figure 9: **Topographic map structure.** Left: displacement vector field showing, for each unit, an arrow from its grid position to a position displaced proportionally to its RF center offset. All arrows converge toward the center of the grid, indicating that the RF gradient is centripetal—peripheral grid units prefer spatial positions slightly offset from the center, and the direction of offset is systematic. Right: RF centers plotted in stimulus space (preferred x vs. preferred y , axes $9.536\text{--}9.544 \times 9.494\text{--}9.504$), with grid-adjacent units connected by lines and colored by grid column (red to blue). The connected map preserves neighborhood relationships: units in adjacent columns and rows are proximate in RF space, forming a compressed but topologically coherent map.

The displacement field (left panel) is particularly informative: the centripetal pattern of arrows shows that all units’ RF centers are drawn toward the image center ($\approx 9.5, 9.5$ pixels), but the *direction* of the offset from center is systematic, creating a compressed but coherent spatial map. The RF center scatter (right panel) confirms this: grid-adjacent units (connected by lines) generally cluster together, and the color gradient (encoding grid column) shows a smooth diagonal progression, demonstrating that the 2D grid topology is preserved in the RF center space despite the extreme compression of the spatial range.

3.5 Permutation Test: Statistical Significance

To assess whether the observed smoothness could arise by chance, we perform a permutation test: the mapping between grid positions and RF centers is randomly shuffled 500 times, and the Spearman ρ is recomputed for each permutation. Figure 10 shows the result.

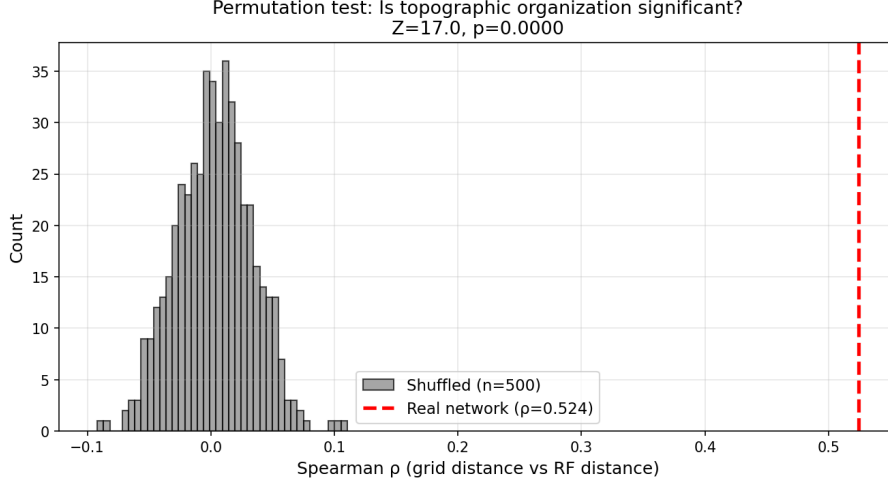


Figure 10: **Permutation test for topographic significance (softmax readout)**. Histogram of Spearman ρ values from 500 random permutations of the grid-to-RF mapping (gray bars), with the observed value $\rho = 0.524$ shown as a red dashed line. The shuffled distribution is centered at $\rho \approx 0.000$ with standard deviation ≈ 0.031 , yielding $Z = 17.0$, $p < 0.0001$. The observed smoothness lies far outside the shuffled distribution, confirming that the spatial organization is not an artifact of the metric or grid geometry.

The result is unambiguous: the observed $\rho = 0.524$ lies 17 standard deviations above the mean of the shuffled distribution ($\mu_{\text{shuf}} \approx 0.000$, $\sigma_{\text{shuf}} \approx 0.031$). No single permutation out of 500 produced a value even approaching the observed correlation, yielding $p < 0.002$ by direct count and $p < 10^{-64}$ by Gaussian approximation.

3.5.1 Control: Hard WTA Permutation Test

As a control, Figure 11 shows the same permutation test applied to the hard WTA readout, which operates on only ~ 4 active units.

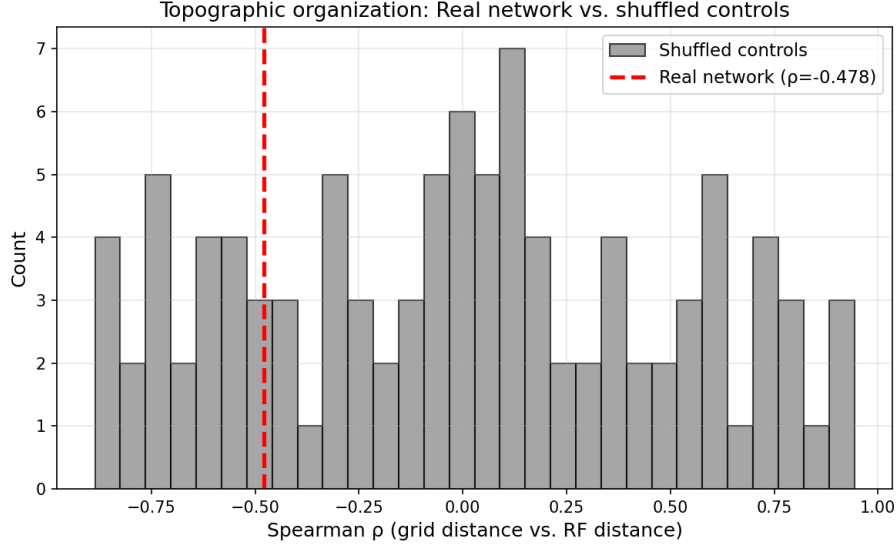


Figure 11: **Control: permutation test under hard WTA.** With only ~ 4 active BMUs, the effective number of RF centers is too small to yield reliable pairwise statistics. The shuffled distribution (gray bars) spans nearly the full range $[-1, +1]$, approximating a uniform distribution rather than a narrow Gaussian. The observed value ($\rho = -0.478$, red dashed line) falls well within the shuffled distribution, demonstrating that the hard WTA readout cannot detect statistically significant spatial organization—the small number of active units produces unstable correlation estimates with effectively no statistical power.

This control underscores the critical importance of the readout method: the same network that produces $Z = 17.0$ under soft readout yields a non-significant, even negative correlation under hard WTA, because collapsing the response to a single winner among ~ 4 active units destroys the graded distance information that carries the spatial signal.

3.6 Receptive Field Tiling

Figure 12 shows how the network tiles the input space under softmax readout, plotting each probe stimulus at its known spatial position and coloring it by the identity of the most strongly activated unit.

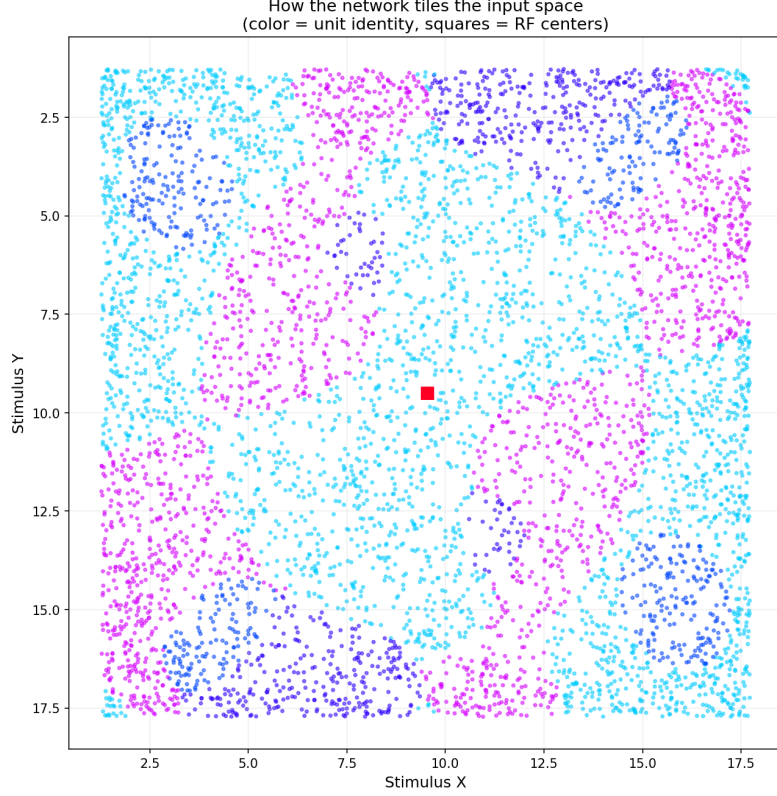


Figure 12: **Receptive field tiling of the input space.** Each of the 20,400 probe stimuli is plotted at its true spatial position (μ_x, μ_y) and colored by the identity of the unit with the highest softmax activation. The color field shows coherent spatial regions associated with individual units, demonstrating that the network partitions the input space into spatially contiguous receptive fields. Square markers indicate RF centers. The tiling covers the full 20×20 pixel extent, confirming that the network’s spatial selectivity, while weak in absolute terms, is spatially structured and covers the input space.

3.7 Effect of Probe Stimulus Width σ

The width of the Gaussian probe stimulus (σ_{stim}) is a free parameter of the analysis. We systematically sweep σ_{stim} from 0.5 to 20 pixels and recompute all topographic metrics at each value. Figure 13 presents the complete results across six panels.

Effect of Gaussian σ on topographic signal detection

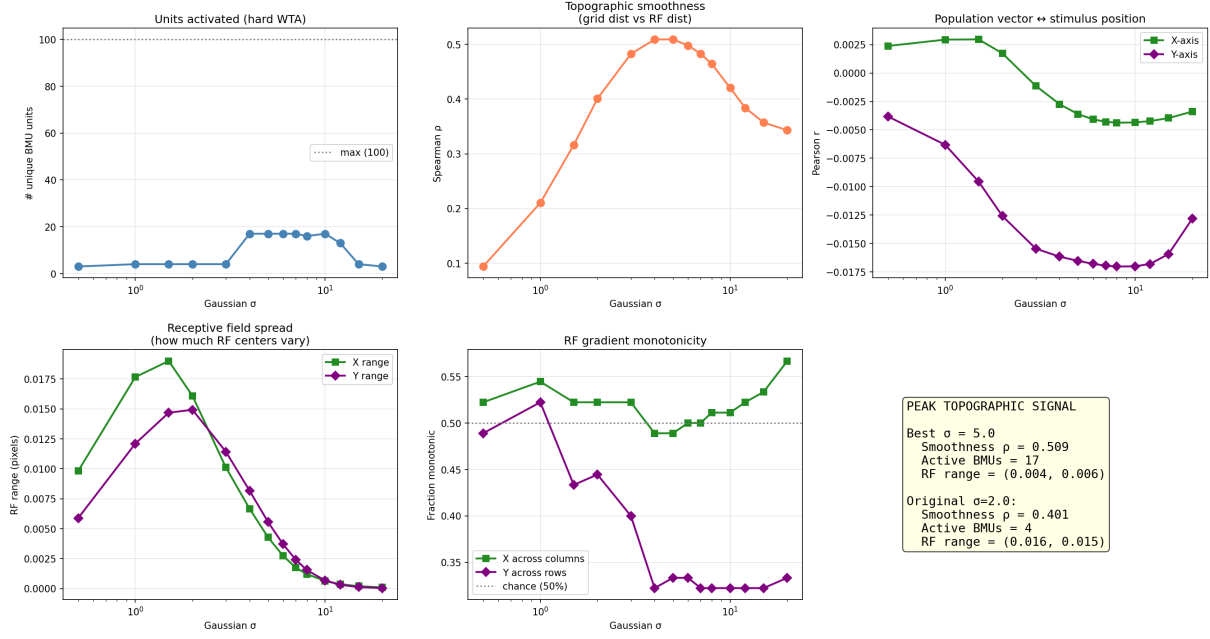


Figure 13: **Effect of Gaussian σ on topographic signal detection.** Six panels show how key metrics vary with probe stimulus width σ_{stim} (log scale, 0.5 to 20 pixels). **Top left:** Number of unique BMUs activated (hard WTA)—ranges from ~ 4 at narrow σ to ~ 17 at intermediate σ , far below the maximum of 100, then drops back at large σ . **Top center:** Topographic smoothness (Spearman ρ)—the inverted-U relationship, peaking at $\rho = 0.509$ at $\sigma = 5.0$ and declining toward ≈ 0.10 at the extremes. **Top right:** Population vector vs. stimulus position correlations (Pearson r)—X-axis correlation near zero throughout; Y-axis correlation peaks at $r \approx -0.017$ around $\sigma = 3-5$. **Bottom left:** RF range (spread of RF centers in pixels)—peaks at $\sigma \approx 1.5-2.0$ with X-range ≈ 0.019 and Y-range ≈ 0.015 , then decreases to near zero at large σ . **Bottom center:** RF gradient monotonicity (fraction of adjacent pairs with consistent RF gradient direction)—peaks slightly above chance (0.50) at narrow σ , near chance at broad σ . **Bottom right:** Summary box: peak topographic signal at $\sigma = 5.0$ with $\rho = 0.509$, 17 active BMUs, RF range (0.004, 0.006); original $\sigma = 2.0$ gives $\rho = 0.401$, 4 active BMUs, RF range (0.016, 0.015).

The σ sweep reveals an *inverted-U* dependence of topographic smoothness on probe width, explained by two competing effects:

1. **Very narrow probes ($\sigma < 1$):** The Gaussian peaks are so localized that they fall within sub-pixel regions of the input space. The response is dominated by the nearest pixel values, making the stimulus undiscriminating—many units respond similarly, and only ~ 4 distinct BMUs are activated. Smoothness is low ($\rho \approx 0.10$).
2. **Optimal range ($\sigma \approx 4-5$):** The probe is broad enough to engage multiple prototypes differentially (17 active BMUs), producing discriminating population responses that expose the map's spatial selectivity. Smoothness peaks at $\rho = 0.509$.
3. **Very broad probes ($\sigma > 10$):** The Gaussian covers most of the 20×20 image, becoming spatially uniform. All probes produce similar network responses regardless of center position, and spatial information is lost. Smoothness drops below 0.10.

An important dissociation emerges between spatial resolution and spatial smoothness. The RF range (bottom left panel) peaks at $\sigma \approx 1.5-2.0$, *before* the smoothness peak. This means the optimal probe for maximizing the spatial *extent* of RF centers differs from the optimal probe for maximizing the spatial *ordering* (smoothness). Narrow probes produce wider RF spreads but noisier orderings; broader probes produce narrower RF spreads but more reliable orderings.

3.8 RBF vs. Softmax Comparison

We compare the unnormalized RBF readout (Eq. 9) against softmax-normalized population coding and hard WTA. Figure 14 presents a comprehensive comparison across six metric panels and a summary table.

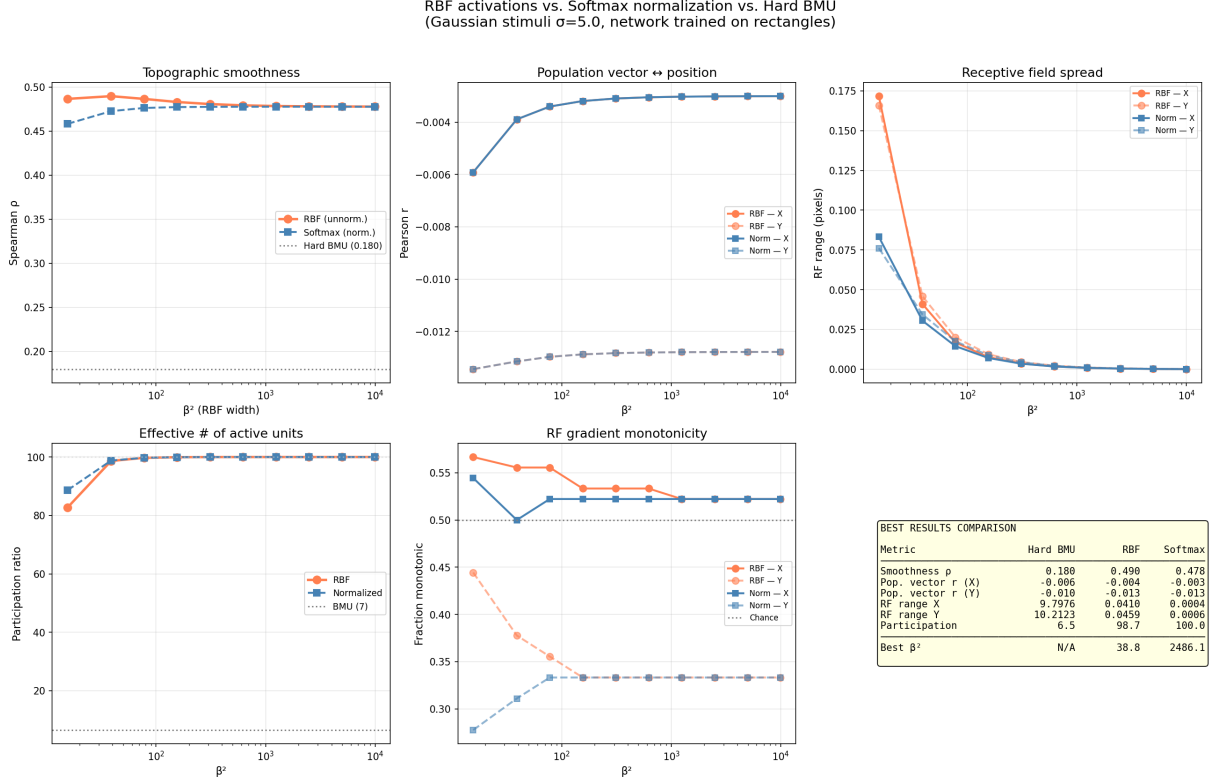


Figure 14: **RBF activations vs. softmax normalization vs. hard BMU.** Gaussian probe stimuli at $\sigma = 5.0$, network trained on rectangles. Six panels show metric variation with β^2 (RBF width parameter, log scale). **Top left:** Topographic smoothness ρ —RBF peaks at $\rho = 0.490$, softmax at $\rho = 0.478$; both far exceed hard WTA at $\rho = 0.180$ (dashed line). **Top center:** Population vector $\leftrightarrow$ stimulus position (Pearson r)—both methods reach $r \approx -0.013$ for y -axis at large β^2 . **Top right:** RF range—RBF produces much larger RF spreads at small β^2 (X-range up to 0.17 pixels) than softmax (peaking at ≈ 0.08), converging at large β^2 . **Bottom left:** Effective number of active units (participation ratio)—RBF ranges from ~ 83 to ~ 100 ; softmax from ~ 89 to 100 ; hard WTA is at 6.5 (dashed line). **Bottom center:** RF gradient monotonicity—RBF exceeds chance at small β^2 (≈ 0.58); softmax remains near chance. **Bottom right:** Summary table of best results: Hard BMU $\rho = 0.180$, RBF $\rho = 0.490$ (best $\beta^2 = 38.8$), Softmax $\rho = 0.478$ (best $\beta^2 = 2486.1$).

The central finding is that RBF and softmax produce nearly equivalent topographic smoothness ($\rho = 0.490$ vs. 0.478), and both vastly outperform hard WTA ($\rho = 0.180$). This near-equivalence has a clear theoretical explanation: the topographic information resides in the *ordering* of distances between the probe and the prototypes, not in the absolute magnitudes. Both softmax and RBF are monotonic functions of distance and therefore preserve this ordering. The normalization in softmax controls the sharpness of the population response but does not change the ranking of unit activations.

One notable difference: at small β^2 values, RBF produces larger RF spreads (up to 0.17 pixels vs. 0.08 for softmax) and higher gradient monotonicity (≈ 0.58 vs. ~ 0.50). This is because unnormalized RBF activations allow the total activation magnitude to vary across stimuli, providing an additional spatial signal absent in the normalized softmax representation.

3.8.1 RF Analysis under RBF Readout

Figure 15 shows the receptive field analysis under the optimal RBF readout ($\beta^2 = 38.8$).

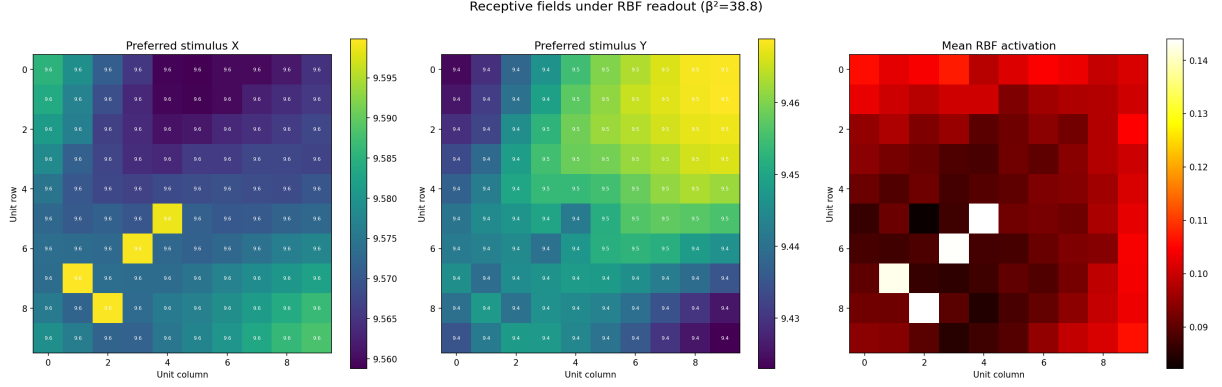


Figure 15: **Receptive fields under RBF readout** ($\beta^2 = 38.8$). Left: preferred stimulus x -coordinate per unit (range ≈ 9.560 – 9.595). Center: preferred stimulus y -coordinate (range ≈ 9.43 – 9.47). Right: mean RBF activation per unit (range ≈ 0.09 – 0.14). Smooth spatial gradients are visible across the grid in both x and y coordinates. The RF range under RBF (≈ 0.04 pixels in x , ≈ 0.04 in y) is larger than under softmax (≈ 0.007 pixels), consistent with the unnormalized activations providing greater spatial discriminability. Mean activation varies across units, with peripheral units showing lower values.

Figure 16 presents the 2D map of RF centers under RBF readout.

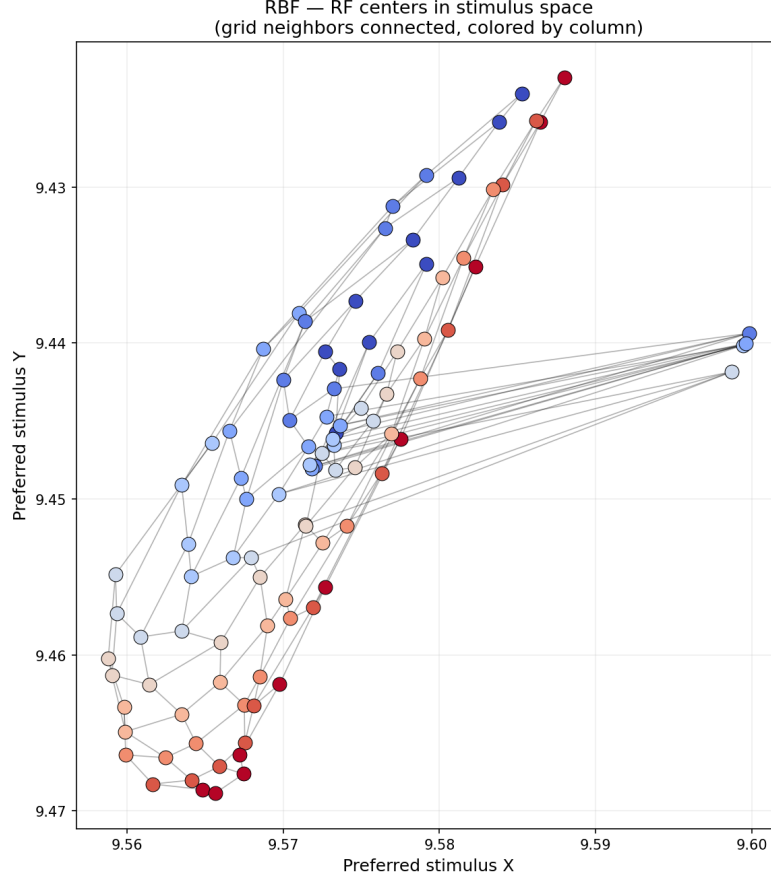


Figure 16: **RBF readout—RF centers in stimulus space.** Preferred stimulus x (range 9.56–9.60) vs. preferred stimulus y (range 9.43–9.47) for all 100 units, with grid-adjacent units connected by lines and colored by grid column (red to blue). The map shows a coherent, elongated diagonal structure preserving neighborhood relationships. The RF spread under RBF ($\approx 0.04 \times 0.04$ pixels) is approximately $5\times$ larger than under softmax ($\approx 0.008 \times 0.010$), but the topological organization is qualitatively identical.

3.8.2 Population Vector under RBF

Figure 17 shows the population vector readout under RBF activations.

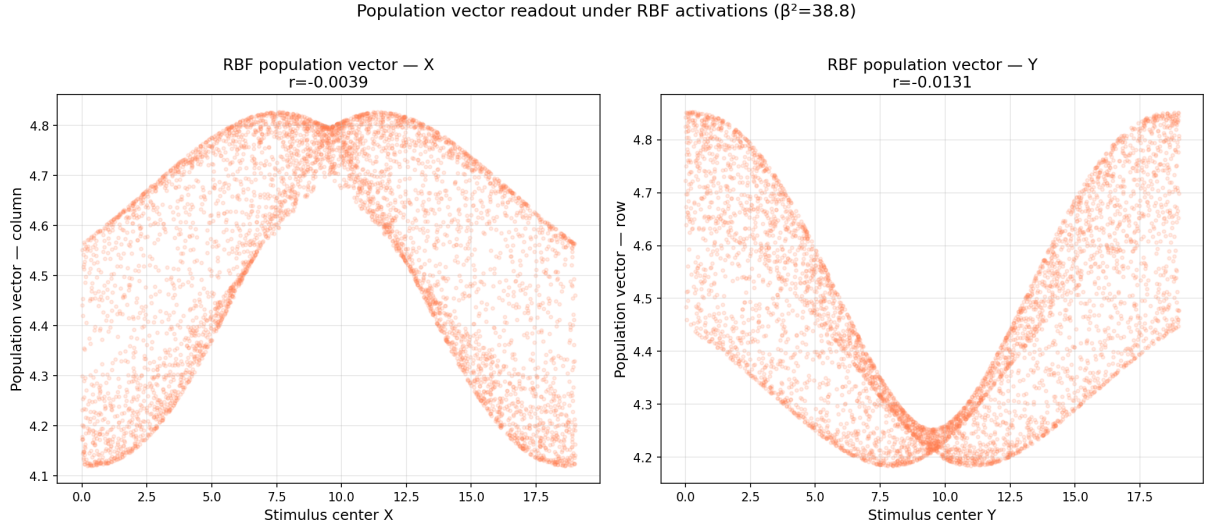


Figure 17: **Population vector readout under RBF activations** ($\beta^2 = 38.8$). Left: RBF population vector column vs. stimulus center x ($r = -0.0039$). Right: RBF population vector row vs. stimulus center y ($r = -0.0131$). Both panels show smooth sinusoidal modulation of the population vector as a function of stimulus position, qualitatively identical to the softmax readout (Figure 6, bottom row). The population vector oscillates between approximately 4.1 and 4.85 grid units, confirming that the spatial signal modulates the population response systematically.

3.8.3 RBF Permutation Test

Figure 18 confirms statistical significance under RBF readout.

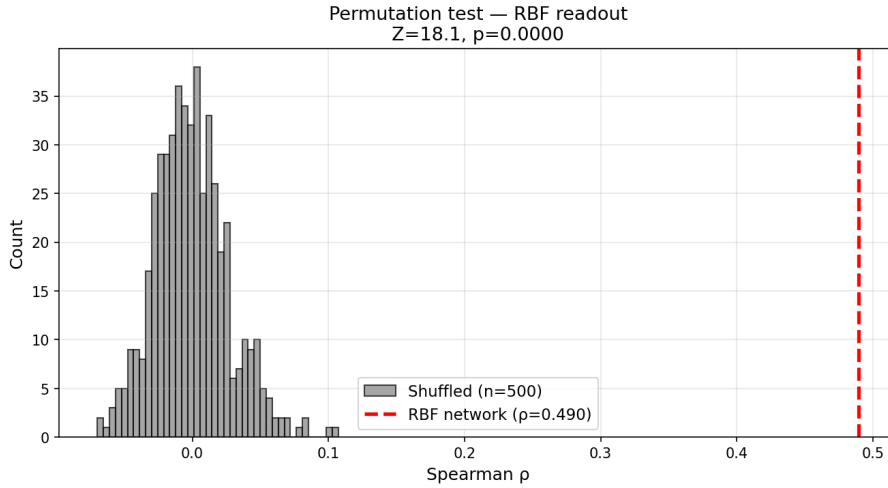


Figure 18: **Permutation test under RBF readout**. Histogram of Spearman ρ from 500 random permutations (gray bars) with the observed RBF smoothness $\rho = 0.490$ (red dashed line). $Z = 18.1$, $p < 0.0001$. The result is even more significant than the softmax permutation test ($Z = 17.0$), confirming that the topographic signal is robust across readout methods.

4 Discussion

4.1 Does a Shape-Trained SOM Acquire the Topology of Its Input?

The central question motivating this analysis is whether a topographic map trained on shape features acquires the *local topology* of its input space. The answer is nuanced.

The SOM is a topology-preserving mapping: it learns to represent the input manifold such that neighborhood relationships in input space are reflected in neighborhood relationships on the grid (Kohonen, 2001). What determines the map’s large-scale organization is the dominant variance in the training data. In our case, shape parameters (aspect ratio, orientation, size, color) contribute far more variance than spatial position, so the global organization of the grid reflects the shape manifold, not a retinotopic map.

However, spatial position is an *embedded dimension* of the shape manifold. Every rectangle occupies a position on the retina, and two rectangles that are identical except for a small shift in position are nearby in input space. The topographic constraint of the SOM ensures that this small-scale spatial structure is preserved locally: units that are grid neighbors tend to encode shapes at similar retinal positions, even though the global map is organized by shape identity.

This is the key distinction: the SOM acquires the topology of the *shape manifold*, which includes spatial position as one of its dimensions. It does *not* acquire a dedicated spatial map in the sense of V1 retinotopy. The spatial structure we detect is residual, local, and embedded within a higher-dimensional organization—but it is real, statistically overwhelming ($Z = 17.0$ – 18.1 , $p < 0.0001$), and detectable with the right methods.

4.2 The Readout Problem: Hard WTA as an Inadequate Decoder

Our most practically significant finding is that hard WTA readout is wholly inadequate for detecting spatial organization in a map trained on non-spatial features. This has direct methodological implications for both computational neuroscience and neurophysiology.

In the SOM, the Gaussian probe stimuli are “off-manifold”—they do not resemble the colored rectangles on which the map was trained. The distances from a probe to all prototypes are large and similar, so the BMU selection is unstable: small changes in the probe can flip the winner to a distant grid location. Hard WTA, by collapsing the population response to a single unit, discards the graded distance information that carries the spatial signal.

Soft readout methods rescue the signal by leveraging the *full* population response. Even though each unit’s spatial preference is weak in absolute terms, the consistent *ordering* of preferences across the population produces a robust spatial signal when combined into a population vector. This is precisely the advantage of population coding over labeled-line coding (Georgopoulos et al., 1986; Averbeck et al., 2006).

The participation ratio quantifies this effect: under soft readout, ~ 83 – 100 units contribute to each prediction (participation ratio 83–100), compared to an effective participation ratio of 6.5 under hard WTA. The effective dimensionality of the population response determines how much spatial information can be recovered.

4.3 Relation to Parietal Retinotopy

Our computational findings parallel a striking observation in neurophysiology: retinotopic organization persists in posterior parietal areas (IPS0–IPS4) that are primarily engaged in spatial attention, motor intention, and object-directed action (Swisher et al., 2007; Silver and Kastner, 2009). These areas exhibit coarse but reliable retinotopic maps, with receptive fields that are progressively larger and more feature-selective as one ascends the hierarchy from V1 (Arcaro et al., 2015). The parietal retinotopic signal is weaker than in early visual cortex but robust at the population level—precisely the regime our SOM analysis models.

Four parallels between our computational results and parietal neurophysiology are particularly informative:

1. **Weak but real spatial signal.** In our SOM, the absolute range of RF centers is small (less than 0.01 pixels out of 20 under softmax; ≈ 0.04 pixels under RBF), yet the spatial smoothness is overwhelmingly significant ($Z = 17.0$ – 18.1). In parietal cortex, individual

neurons have large, feature-dominated receptive fields, yet population-level analyses reveal reliable spatial maps (Serenio et al., 2001).

2. **Readout method matters.** Hard WTA readout fails to detect spatial structure in our SOM ($\rho = 0.180$), whereas population coding succeeds ($\rho = 0.478\text{--}0.524$). Analogously, single-neuron recordings in parietal cortex may miss spatial selectivity that population-level analyses can recover (Serenio et al., 2001). This is not a limitation of the map but of the readout.
3. **Optimal probe scale.** The inverted-U dependence on σ shows that the topographic signal is maximized when the probe matches the spatial scale of the map’s receptive fields (peak at $\sigma = 5.0$, $\rho = 0.509$). In neurophysiology, the spatial structure of parietal maps is best revealed by stimuli at an appropriate spatial scale—neither point stimuli (too narrow for the large receptive fields) nor full-field stimuli (too broad to be spatially discriminating) (Wandell et al., 2007).
4. **Spatial scaffold for non-spatial organization.** Our SOM is organized around shape features but retains residual spatial structure as a geometric byproduct of topographic learning on spatially structured input. Parietal areas are organized around task-relevant features (attention priority, motor plans) but retain retinotopic structure because their inputs are spatially structured (Arcaro et al., 2015).

This perspective offers a reinterpretation of parietal retinotopy. The conventional view treats retinotopic maps in IPS as evidence for dedicated spatial coordinate frames that are predictively updated by corollary discharge signals during saccades (Duhamel et al., 1992; Sommer and Wurtz, 2006). Our analysis suggests an alternative: the spatial signal in parietal cortex may reflect *residual local retinotopy*—a geometric byproduct of topographic learning—rather than a dedicated coordinate system. The “remapping” observed in such areas may be better understood as a consequence of topographic continuity than as evidence for an explicit coordinate transformation.

This does not diminish the functional significance of parietal spatial signals. On the contrary, our analysis reveals that residual spatial structure is not a vestige but serves a crucial computational role, as we argue next.

4.4 Implications for Topological Alignment

The topological alignment framework (Mannella and Tummolini, 2023) proposes that multiple sensory and motor modalities become grounded through coordinated topographic self-organization within a shared internal space. In this framework, supervised topological maps (STMs) for vision, somatosensation, proprioception, and motor policy are co-trained on a common grid, constrained by a shared anchor mechanism that forces corresponding representations across modalities to converge to the same grid location.

A critical requirement for this alignment to succeed is that the individual maps must share *enough common structure* for the anchor mechanism to find correspondences. For a visual map organized around object features to align with the inherently spatial maps of proprioception and touch, the visual map must retain some spatial information. If a feature-organized visual map completely lost spatial structure, it would have nothing in common with the spatial maps of other modalities, and alignment would fail.

Our SOM analysis demonstrates that this requirement is met automatically: a topographic map trained on spatially structured visual input *inevitably* retains local spatial organization, regardless of what other features dominate its global structure. The residual retinotopy we detect is precisely the spatial scaffold that makes cross-modal alignment possible.

This yields the central hypothesis of this report: *residual spatial structure in feature-organized topographic maps is not a nuisance or a vestige, but a necessary geometric property that enables cross-modal alignment*. Parietal areas are not retinotopic *despite* processing shapes and actions; they retain spatial structure *because* the topology of visual input, from which shapes and actions

are learned, is itself spatial. The coarse retinotopy of IPS0–IPS4 is the geometric bridge that allows visual, somatosensory, proprioceptive, and motor representations to be aligned within a common internal space.

This perspective connects our computational findings to a broader theoretical framework. The Mannella & Tummolini framework identifies the shared internal space as the locus of semantic grounding: a point in the internal space “refers to” an environmental condition when executing the associated policy reliably produces the associated sensory consequences. The spatial scaffold provided by residual retinotopy is what enables this grounding to extend across modalities—it is the geometric prerequisite for the triadic structure (condition–action–effect) that defines goal-directed behavior in the framework.

Recent evidence of pervasive retinotopic biases in areas traditionally considered non-retinotopic—including inferotemporal cortex, scene-selective regions, and face patches (Arcaro and Livingstone, 2017)—is consistent with this view. Arcaro et al. (2015) have argued that innate retinotopic biases may guide the development of category-selective cortical organization, a proposal that complements our framework: retinotopy provides the initial spatial scaffold on which feature-selective organization is built, and the residual spatial structure that persists after feature learning enables the cross-modal alignment that grounds perception in action.

5 Conclusion

We have demonstrated that a self-organizing map trained on a high-dimensional shape manifold retains statistically overwhelming local spatial organization, detectable only through soft population readout methods. Our analysis yields four principal findings:

1. **Residual spatial structure is real and robust.** The spatial smoothness of the shape-trained SOM is highly significant ($\rho = 0.524$, $Z = 17.0$, $p < 0.0001$ under softmax; $\rho = 0.490$, $Z = 18.1$, $p < 0.0001$ under RBF), despite the map being organized primarily around shape features.
2. **The readout method is critical.** Hard WTA readout fails to detect spatial structure ($\rho = 0.180$, activating only ~ 4 distinct BMUs across 20,400 probes); soft population coding (softmax or RBF) recovers the signal by leveraging the full population response (participation ratio 83–100). This has direct implications for neurophysiological studies of spatial organization in association cortex.
3. **Probe stimulus width must match the map’s spatial scale.** The topographic signal follows an inverted-U as a function of σ_{stim} , peaking at $\sigma = 5.0$ ($\rho = 0.509$, 17 active BMUs). The dissociation between spatial resolution (RF range peaks at $\sigma \approx 1.5$ –2.0) and spatial smoothness (peaks at $\sigma \approx 5.0$) reflects competing demands of discriminability and coverage.
4. **Residual retinotopy serves as a spatial scaffold for cross-modal alignment.** Within the topological alignment framework (Mannella and Tummolini, 2023), the local spatial structure retained by feature-organized maps is a necessary geometric condition for aligning visual, somatosensory, proprioceptive, and motor representations within a shared internal space.

These findings suggest that the observation of coarse retinotopic maps in parietal cortex (IPS0–IPS4) should not be interpreted as evidence for dedicated spatial coordinate frames in the traditional sense. Rather, local retinotopy is an intrinsic geometric property of any topographic representation trained on spatially structured visual input, and it serves the crucial functional role of enabling cross-modal manifold alignment for goal-directed action control.

A SOM Training Algorithm

Algorithm 1 Batch SOM Training

Require: Dataset $\{\mathbf{x}_n\}_{n=1}^N$, grid positions $\{\mathbf{p}_j\}_{j=1}^K$

Require: Hyperparameters: $\eta_0, \sigma_0, T, \tau, B$

```

1: Initialize  $\mathbf{W} \in \mathbb{R}^{K \times M}$  randomly
2: Precompute grid distance matrix:  $G_{jk} = \|\mathbf{p}_j - \mathbf{p}_k\|^2$ 
3: for  $t = 1, \dots, T$  do
4:    $\eta \leftarrow \eta_0 \exp(-t/\tau)$ 
5:    $\sigma^2 \leftarrow [\sigma_0 \exp(-t/\tau)]^2$ 
6:   for each mini-batch  $\mathcal{B}$  of size  $B$  do
7:      $\|\mathbf{x}\|^2 \leftarrow$  row norms of batch  $\triangleright B$ -vector
8:      $\|\mathbf{w}\|^2 \leftarrow$  row norms of  $\mathbf{W}$   $\triangleright K$ -vector
9:      $D^2 \leftarrow \|\mathbf{x}\|^2 \mathbf{1}^T + \mathbf{1} \|\mathbf{w}\|^2 - 2 \mathcal{B} \mathbf{W}^T$   $\triangleright B \times K$ 
10:     $\mathbf{r} \leftarrow \arg \min_j D_{n,j}^2$  for each  $n$   $\triangleright$  BMU indices
11:     $\Phi_{n,j} \leftarrow \exp(-G_{r_n,j}/2\sigma^2)$   $\triangleright B \times K$ 
12:     $\mathbf{S} \leftarrow \Phi^T \mathcal{B}$   $\triangleright K \times M$  weighted input sums
13:     $\mathbf{Z} \leftarrow \sum_n \Phi_{n,j}$   $\triangleright K$ -vector of normalization
14:     $\mathbf{W} \leftarrow \mathbf{W} + \eta (\mathbf{S} \oslash (\mathbf{Z} + \epsilon) - \mathbf{W})$ 
15:   end for
16: end for

```

B Complete Metric Formulas

Population Vector

Given activation vector $\mathbf{a} = (a_1, \dots, a_K)$ and grid positions $\{\mathbf{p}_j\}$, the population vector is:

$$\hat{\mathbf{p}} = \frac{\sum_j a_j \mathbf{p}_j}{\sum_j a_j} \quad (15)$$

RF Center

The receptive field center of unit j , computed over all probe stimuli:

$$\mathbf{c}_j = \frac{\sum_{n=1}^{N_{\text{probe}}} a_j(\mathbf{x}_n) \boldsymbol{\mu}_n}{\sum_{n=1}^{N_{\text{probe}}} a_j(\mathbf{x}_n)} \quad (16)$$

Smoothness (Spearman ρ)

$$\rho = 1 - \frac{6 \sum_i d_i^2}{n(n^2 - 1)} \quad (17)$$

where d_i is the difference in ranks of the i -th pair and $n = \binom{K}{2}$.

Participation Ratio

$$\text{PR}(\mathbf{a}) = \frac{\left(\sum_{j=1}^K a_j\right)^2}{\sum_{j=1}^K a_j^2} \quad (18)$$

RF Gradient Monotonicity

For unit j , consider its grid neighbors $\mathcal{N}(j)$. The gradient monotonicity measures whether RF centers change monotonically along grid axes:

$$M_j = \frac{1}{|\mathcal{N}(j)|} \sum_{k \in \mathcal{N}(j)} \text{sign}(\langle \mathbf{c}_k - \mathbf{c}_j, \mathbf{p}_k - \mathbf{p}_j \rangle) \quad (19)$$

Adaptive Temperature (Softmax)

$$\tau^2 = 2 \cdot \text{median} \left(\left\{ \min_j \|\mathbf{x}_n - \mathbf{w}_j\|^2 \right\}_{n=1}^{N_{\text{probe}}} \right) \quad (20)$$

C Summary of Numerical Results

Table 1: Comparison of readout methods. RBF results at optimal $\beta^2 = 38.8$; softmax at optimal $\beta^2 = 2486.1$. Gaussian probes at $\sigma_{\text{stim}} = 5.0$.

Metric	Hard WTA	Softmax	RBF
Spatial smoothness ρ	0.180	0.478	0.490
Z-score (permutation)	n.s.	17.0	18.1
p -value	n.s.	<0.0001	<0.0001
Pop. vector $r(x)$	-0.006	-0.003	-0.004
Pop. vector $r(y)$	-0.010	-0.013	-0.013
RF range x (pixels)	9.798	0.0004	0.0410
RF range y (pixels)	10.212	0.0006	0.0459
Participation ratio	6.5	100.0	98.7
Optimal β^2	N/A	2486.1	38.8

Table 2: Effect of probe stimulus width σ_{stim} (softmax readout).

Property	Optimal σ	Value
Peak smoothness ρ	5.0	0.509
Active BMUs at peak	5.0	17
RF range at peak	5.0	(0.004, 0.006)
Peak RF range	1.5–2.0	(0.019, 0.015)
Original $\sigma = 2.0$: smoothness	2.0	0.401
Original $\sigma = 2.0$: active BMUs	2.0	4
Original $\sigma = 2.0$: RF range	2.0	(0.016, 0.015)
Smoothness at $\sigma = 0.5$	0.5	≈ 0.10
Smoothness at $\sigma = 20$	20	≈ 0.10

Table 3: Permutation test results ($n_{\text{perm}} = 500$).

Readout	Observed ρ	Z-score	p-value
Softmax ($\sigma_{\text{stim}} = 2.0$)	0.524	17.0	<0.0001
RBF ($\sigma_{\text{stim}} = 5.0$, $\beta^2 = 38.8$)	0.490	18.1	<0.0001
Hard WTA (control)	-0.478	n.s.	n.s.

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